Westside Stylist Mirror: what the purchase data says

ANCP D3 · EDA insight summary · Group M · C Prashanth · Linked outcomes CO2, CO4 · Data: H&M 10% customer sample (3,205,663 rows, Sep 2018–Sep 2020) as a proxy for Westside purchase history

Bottom line

The data support building the mirror as an **outfit completer and re-buy assistant**, not as a taste-learner for one-item shoppers. Fewer than half of garment visits (45%) contain a complete outfit, but baskets are not thin on average (30% single-unit visits; 3.5 units per visit). History helps mainly through exact re-buys, and demand turns over within weeks. The 10% sample matches the full file on every representativeness check, so the main risks are the proxy (European, no gender or store field) and the 2020 break, not sampling.

The five insights that matter most, and how each shapes modelling

Insight	Evidence	Shapes modelling
1 The gap is the incomplete outfit, not the one-item basket.	Garment visits: complete 45%, top only 38%, bottom only 17%. Single-unit visits 30% (95% CI 30%–30%); thin baskets sit in one channel (38% vs 25%; meaning of the code unverified).	Set the items-per-bill baseline at about 3.5, not 1, and target top-only / bottom-only moments. Revise uplift assumptions in the M6 simulation.
2 History predicts re-buying, not taste.	Exact-article repeat: 5.4% vs 0.3% for a same-week random draw (19×). Product type 1.26×, colour 1.07×.	M4 recommender: start with repeat and recency features and item-to-item cross-sell; add taste embeddings only if they beat that.
3 Demand turns over within weeks.	Last week's top 12 covers 1.6% of next week's units, top 100 covers 8.4%; top-100 overlap 0.29 at 1 week (noise ceiling 0.37), 0.13 at 4 weeks. 39% of weekly units come from articles first sold ≤ 8 weeks earlier.	Rebuild the best-seller baseline weekly; validate by date; give new articles an attribute-based fallback; add season terms to the forecast.
4 Heavy buyers dominate; many shoppers have no history; older shoppers buy least per visit.	Top 10% of customers buy 48% of units; 32% have one visit. Single-unit share by age: 65+ 43% vs 25–34 27%.	Split by customer, report by activity tier and age band, weight heavy buyers, and ship a popularity / content fallback for one-visit members.
5 Co-purchase cannot identify complements; colour is mostly neutrals.	Median adjusted lift: top×bottom 1.35, top×top 1.36 (no difference). Black, White, Blue = 62% of units; light share 42% summer vs 30% winter.	Use co-purchase only to propose candidates; train and test the outfit matcher on Polyvore. Test colour advice against shopper-chosen colours in M5.

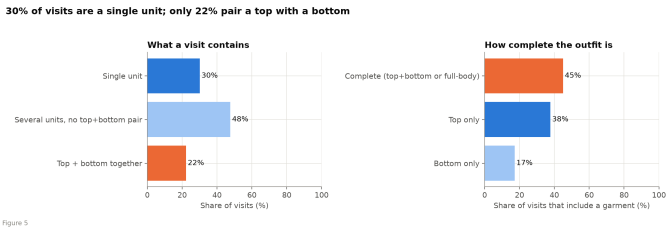


Figure 5. Most visits are not single-unit; outfits are incomplete.

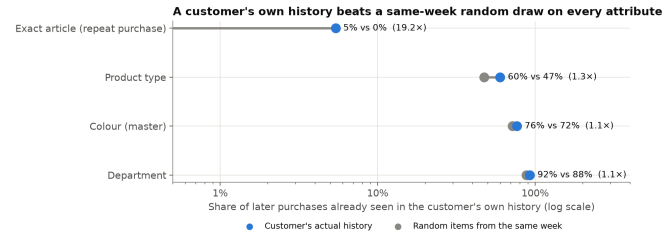


Figure 6. History is strongest for exact re-buys.

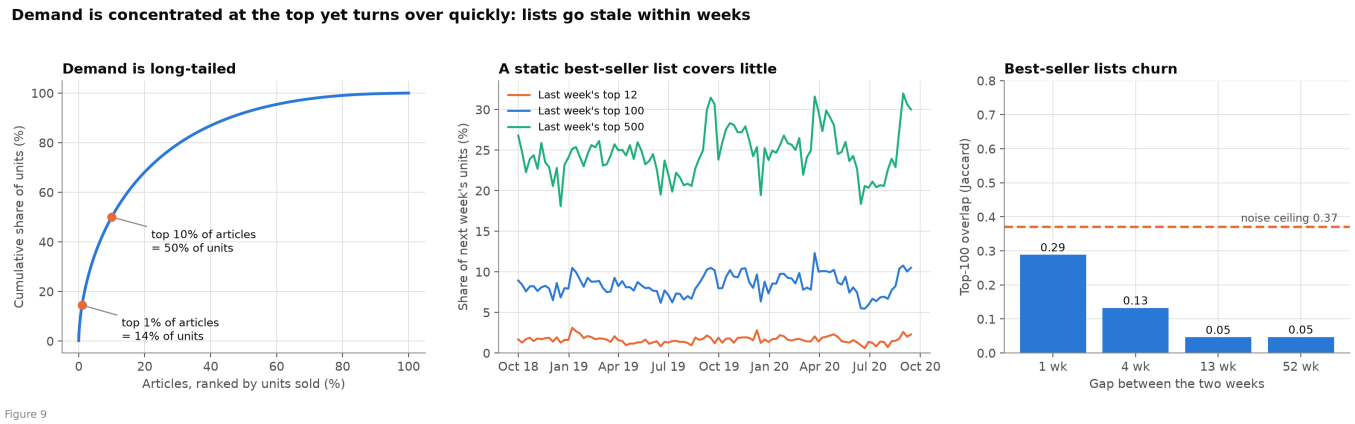


Figure 9. Demand is concentrated at the top and the top list changes quickly.

Hypotheses, bias checks and limits

Seven hypotheses tested against rules fixed before loading the data

ID	Question	Hypothesis	Evidence vs rule	Verdict
H1	BQ1	Thin baskets, outfit rarely complete	Single-unit 30% (rule ≥ 50%); top+bottom 22% (rule ≤ 15%)	**Not supported**
H2	BQ2	History predicts next purchase (≥ 1.5×)	Type 1.26×, colour 1.07×; exact article 19× (not in rule)	**Not supported**
H3	BQ3	Top–bottom pairs are complementary	Median lift 1.35 (rule ≥ 1.2); top×top 1.36, so not specific to complements	**Supported**
H4	BQ4	Long tail, fast churn	Top 1% of articles hold 14% (rule < 25%); overlap 0.29 at 1 wk (ceiling 0.37), 0.13 at 4 wk	**Supported**
H5	BQ5	Segments differ	Cramér's V 0.06 (rule ≥ 0.15) fails; single-unit gap 17 pts across age bands passes	**Not supported**
H6	BQ2/5	Heavy buyers dominate	Top 10% of customers buy 48% of units (rule ≥ 40%); Gini 0.64	**Supported**
H7	BQ3/4	Colour concentrated and seasonal	Top 3 colours 62% (rule ≥ 50%); light share 42% vs 30%	**Supported**

Four supported, three not. The three failures each correct a charter assumption (thin baskets, taste-based history, age-driven department mix).

Representativeness and bias, repeated on raw and cleaned data

Sample vs full population: 6 of 6 checks pass on raw data and the cleaned frame: department mix (largest gap 0.05 pp), monthly profile (0.06 pp), channel mix (0.02 pp), basket size, purchases per customer (23.5 vs 23.3), customer age and club status (largest |SMD| 0.02). **Cleaning:** duplicate rows kept as quantity (296,434 rows, units preserved); Baby/Children (3.8% of units), non-apparel and under-18 customers excluded; 3,066,344 units in scope. **Remaining bias:** missing age, FN flag or club status mark light buyers (median 1–2 visits vs 3–4); units in Mar–Aug 2020 were -13% vs 2019 and channel-2 share rose from 70% to 75%; no gender, size or store field; European assortment; price is normalised.

Insight log (full table with evidence and so-what in the notebook, section 9)

ID	Insight	Decision it informs	Confidence
I1	Baskets are not thin on average; outfits are incomplete.	Mirror scope; items-per-bill baseline	High (H&M; data); Medium for transfer to Westside
I2	Thin baskets sit in one sales channel.	In-store simulation inputs	Medium (channel label unverified)
I3	A customer's own history is a strong predictor only at the level of the exact article.	Recommender features	Medium (ratio depends on history length; measured on H&M; only)
I4	Purchases show which types are bought together, but not which are complements.	Outfit-matcher data source	Medium
I5	Demand turns over fast.	Baseline, forecast, validation design	High (direction); Medium (levels, 10% sample is noisy)
I6	A minority of heavy buyers dominates the data and a large group has almost no history.	Splits, weighting, cold-start	High
I7	Segments differ more in basket thinness than in what they buy.	Fairness plan (M5)	High
I8	Colour choice is concentrated in three neutrals and follows the season.	Colour-advice value; seasonality terms	Medium (colour attribute is the catalogue's, not the shopper's)
I9	The sample is not the bias; the proxy and the period are.	Validation windows; transfer to Westside	High (sample); Medium (cause of the 2020 break)
I10	The 'single-item habit' group the charter targets is visible but small, and transactions cannot show confidence or effort.	Business-case sizing; survey	Low

Confidence: High = large sample, narrow interval, same direction on raw and cleaned data; Medium = depends on an unverified label, noisy measure or Westside transfer; Low = behavioural proxy.

Limits and next steps

H&M customers are not Westside's; every number is a pattern to re-validate on Westside data. The target group (low-effort, low-confidence dressers) cannot be seen in transactions; our proxy (4+ visits, 60%+ single-unit) covers 10% of repeat customers and 3.5% of their units, and the M3 shopper survey must confirm it. Next: confirm what the sales-channel code means; run the survey; carry the weekly baseline, date-and-customer splits and age-band fairness checks into D4/D5.

Code and files: https://github.com/cprashanth666666-sys/Captain/tree/D3